Continuous Random Variables and the Normal Distribution

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LESE152

Continuous Random Variables: Several Concepts

- Continuous Random Variables
- The Uniform Distribution
- The Normal Distribution
- The Exponential Distribution
- Multiple Random Variables
- Practice Questions

Continuous Random Variables

- Continuous random variable: A random variable whose possible values fill an interval, so they cannot be listed.
- Examples: a price, a wage, the amount of rainfall, the time it takes to drive to school, the labor force participation rate (any number between 0% and 100%).
- Each single value has probability zero, so the math uses calculus instead of sums.

Cumulative Distribution Function (CDF)

- The CDF has the same definition as in Lecture 5:

$$F(x) = P(X \leq x).$$

- A discrete CDF is a step function. A continuous CDF increases smoothly: it can have flat parts, but it never jumps.
- Standard uniform: X can take any value in $[0, 1]$, with

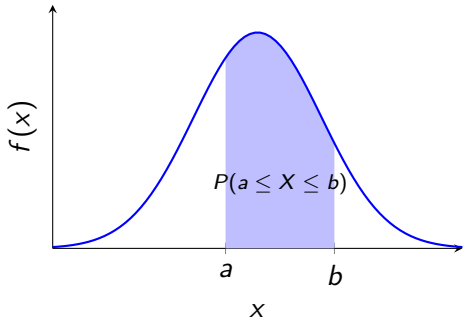
$$F(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

- Interval probabilities work exactly as before:

$$P(0.65 < X < 0.70) = F(0.70) - F(0.65) = 0.05.$$

Probability Density Function (PDF)

- Probability density function: The slope of the CDF,
$$f(x) = F'(x).$$
- Probabilities are areas under the PDF:
$$P(a \leq X \leq b) = \int_a^b f(x)\,dx = F(b) - F(a).$$
- $f(x) \geq 0$, and the total area under f is 1.
- $f(x)$ is a density, not a probability. It can be greater than 1. Only areas are probabilities.



Quantiles and Expected Values

- Quantile: The q quantile x_q solves $F(x_q) = q$. The median is $x_{0.5}$, and the 90th percentile is $x_{0.9}$. These are usually easier to find for continuous random variables.
- Expected value: The same idea as in Lecture 5, with the sum replaced by an integral,

$$E(X) = \int x\, f(x)\, dx.$$

- Variance: $\mathrm{Var}(X) = E[(X-\mu)^2] = E(X^2) - \mu^2$, as before.
- All the rules from Lecture 5 carry over: $E(a+bX) = a + b\,E(X)$, $\mathrm{Var}(a+bX) = b^2\,\mathrm{Var}(X)$, and standardization.
- You will not be asked to integrate. For each standard distribution, we report the mean and variance and use them.

Uniform Distribution

- $X \sim U(a, b)$: all values in $[a, b]$ are equally likely,

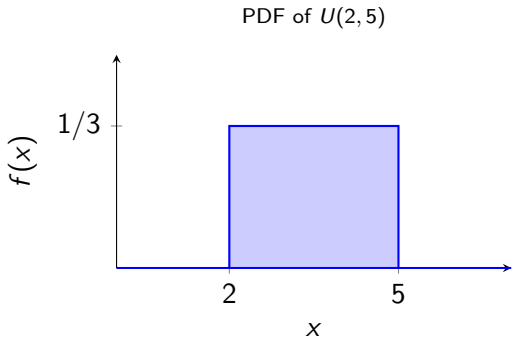
$$f(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b,$$

and $f(x) = 0$ otherwise.

- CDF:

$$F(x) = \frac{x-a}{b-a} \quad \text{for } a \leq x \leq b.$$

- $U(0, 1)$ is called the standard uniform.



Uniform: Quantiles and Moments

- Quantiles: Invert the CDF,
$$x_q = a + q(b - a).$$
The median is the midpoint $(a + b)/2$.
- Mean and variance:
$$E(X) = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$
- $X \sim U(2, 5)$: $E(X) = 3.5$, $\mathrm{Var}(X) = 9/12 = 0.75$, $\sigma \approx 0.866$, and the 90th percentile is $2 + 0.9 \times 3 = 4.7$.
- A linear function of a uniform is uniform: if $X \sim U(2, 5)$, then $2X + 2 \sim U(6, 12)$.

Uniform: An Example

A delivery is equally likely to arrive at any time between 9:00 and 11:00. Let X be the arrival time in minutes after 9:00, so $X \sim U(0, 120)$.

- Arrives before 9:30: $P(X < 30) = 30/120 = 0.25$.
- Arrives exactly at 10:00: $P(X = 60) = 0$. Only intervals have positive probability.
- Expected arrival: $E(X) = 60$, i.e., 10:00. Standard deviation: $120/\sqrt{12} \approx 34.6$ minutes.
- 90th percentile: $0.9 \times 120 = 108$, i.e., 10:48. You can be 90% sure the delivery arrives by then.
- The standard uniform is the building block of simulation: `RAND()` in Excel and `runif()` in R draw from $U(0, 1)$.

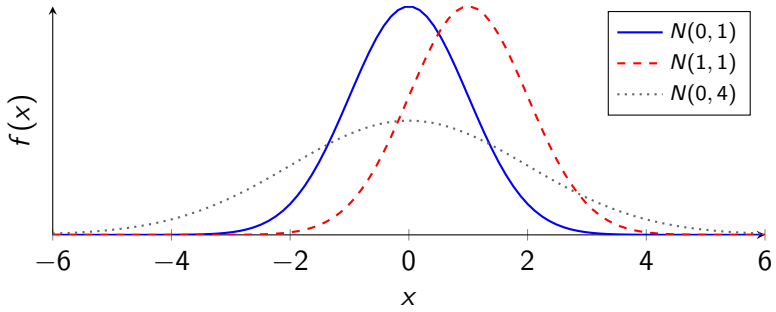
Normal Distribution

- $X \sim N(\mu, \sigma^2)$, with PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty.$$

- Mean and variance: $E(X) = \mu, \quad \mathrm{Var}(X) = \sigma^2$.
- Bell-shaped and symmetric around μ, so mean = median = mode. The total area is 1, with 0.5 on each side of μ.
- Also called the Gaussian distribution. $N(0, 1)$ is the standard normal.
- Why it is everywhere: sums and averages of many small independent shocks are approximately normal.

Normal Shapes



- Increasing μ shifts the curve to the right. Increasing σ flattens and spreads it out.
- Otherwise the shape is identical, so one table (the standard normal) is enough for all normals.

Linear Functions of a Normal

- Any random variable has $E(a+bX) = a + b\,E(X)$ and $\mathrm{Var}(a+bX) = b^2\,\mathrm{Var}(X)$.
- Normal random variables have an extra property: a linear function of a normal is also normal,
$$X \sim N(\mu, \sigma^2) \;\Rightarrow\; a + bX \sim N(a + b\mu,\; b^2\sigma^2).$$
- Example: $X \sim N(2, 9)$. Then
$$X + 1 \sim N(3, 9), \qquad 2X \sim N(4, 36), \qquad -2X \sim N(-4, 36).$$
- Nonlinear functions of a normal are generally not normal. For example, X^2 cannot be negative.

The Standard Normal

- Standardization is especially useful for normals:

$$X \sim N(\mu, \sigma^2) \;\Rightarrow\; Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

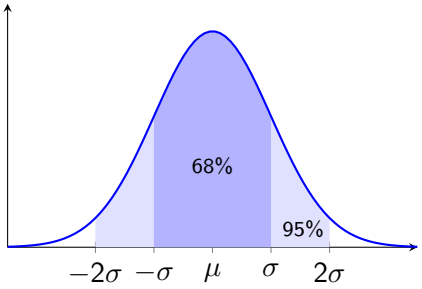
- $\Phi(z) = P(Z \leq z)$ denotes the standard normal CDF.
- Any normal probability can be written with Φ:

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

- The z-score $(x - \mu)/\sigma$ measures how many standard deviations x is from the mean.
- Φ has no simple formula, but it is in every table and every software package.

The Empirical Rule

- For any normal distribution:
  - $P(\mu - \sigma < X < \mu + \sigma) \approx 0.68$
  - $P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95$
  - $P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$
- This is where the 68–95–99.7 rule from Lecture 2 comes from.
- More precise values used throughout the course:
  - $P(|Z| \leq 1.645) = 0.90$
  - $P(|Z| \leq 1.96) = 0.95$
  - $P(|Z| \leq 2.576) = 0.99$



Finding Standard Normal Probabilities

- A standard normal table gives $\Phi(z) = P(Z \leq z)$. Example: $P(Z \leq 1.00) = 0.8413$ (row 1.0, column 0.00).
- Useful rules:
  - Complement: $P(Z > z) = 1 - \Phi(z)$.
  - Symmetry: $P(Z > z) = P(Z < -z)$, so $\Phi(-z) = 1 - \Phi(z)$.
  - Intervals: $P(a \leq Z \leq b) = \Phi(b) - \Phi(a)$.
- $P(-1.5 \leq Z \leq 2.0) = 0.9772 - 0.0668 = 0.9104$.
- $P(Z \geq 1.75) = 1 - 0.9599 = 0.0401$, or equivalently $P(Z \leq -1.75) = 0.0401$.
- Excel: `NORM.S.DIST(z,TRUE)` and `NORM.DIST(x,`μ`,`σ`,TRUE)`. R: `pnorm(z)` and `pnorm(x,mu,sigma)`.
- Sketch the area first. Most errors are set-up errors, not table errors.

Normal Probabilities: Weekly Revenue

A store's weekly revenue is $X \sim N(25000, 4000^2)$.

- Revenue below 20,000:

$$P(X < 20000) = P\left(Z < \frac{20000 - 25000}{4000}\right) = \Phi(-1.25) \approx 0.106.$$

- Revenue above 30,000: $z = 1.25$, so $P(X > 30000) = 1 - \Phi(1.25) \approx 0.106$. The same as below 20,000, by symmetry.
- Revenue between 22,000 and 28,000: $z = \pm 0.75$, so

$$P(22000 < X < 28000) = \Phi(0.75) - \Phi(-0.75) = 0.7734 - 0.2266 \approx 0.547.$$

- Excel: `NORM.DIST(20000,25000,4000,TRUE)`. R: `pnorm(20000,25000,4000)`.

Percentiles: Working Backwards

- Sometimes we know the probability and want the value. The q quantile of $X \sim N(\mu, \sigma^2)$ is

$$x_q = \mu + z_q \, \sigma,$$

  where z_q is the q quantile of the standard normal, $\Phi(z_q) = q$.
- 90th percentile of Z: $\Phi(1.28) \approx 0.90$, so $z_{0.90} \approx 1.28$.
- 90th percentile of $X \sim N(100, 15^2)$: $100 + 1.28 \times 15 = 119.2$.
- The middle 90% of a normal lies within $\mu \pm 1.645 \, \sigma$.
- Excel: `NORM.INV(q,`μ`,`σ`)`. R: `qnorm(q,mu,sigma)`.

Percentiles: Safety Stock

A store's weekly demand is $X \sim N(500, 60^2)$. A delivery arrives once a week, and the current policy stocks 550 units.

- Probability of running out:

$$P(X > 550) = 1 - \Phi\left(\frac{550 - 500}{60}\right) = 1 - \Phi(0.83) \approx 0.20.$$

  The store runs out about once every five weeks.

- To run out only 2% of the time, stock the 98th percentile of demand:

$$x_{0.98} = 500 + 2.054 \times 60 \approx 623 \text{ units.}$$

- The extra 73 units are the cost of reliability. Quantiles translate a service target into an inventory level.

Joint Distribution

- Most data sets have several variables, so we need to describe several random variables at once.
- Joint distribution of X and Y: The probabilities of all pairs of values. For discrete random variables, the joint PMF is

$$p(x, y) = P(X = x \text{ and } Y = y).$$

Marginal and Conditional Distributions

- Marginal distribution: The distribution of one variable alone (the row and column totals in the table):
$$p_X(x) = \sum_y p(x, y).$$
- Roulette: $P(y_{Red} = 1) = 17/37 + 1/37 = 18/37$, the same PMF as in Lecture 5.
- Conditional distribution: The distribution of Y given $X = x$, from Lecture 4's conditional probability,
$$P(Y = y \mid X = x) = \frac{p(x, y)}{p_X(x)}.$$
- Roulette: $P(y_{14} = 35 \mid y_{Red} = 1) = (1/37)/(18/37) = 1/18$.
- Wages given education, yields given rainfall: the conditional mean $E(Y \mid X = x)$ is the population regression function.

Independence

- X and Y are independent if the joint PMF equals the product of the marginals for every pair of values:

$$p(x, y) = p_X(x)\, p_Y(y).$$

  Equivalently, the conditional distribution of Y given X does not depend on X.
- Roulette, same spin: $P(y_{Red} = 1, y_{14} = 35) = 1/37$, but $P(y_{Red} = 1)\, P(y_{14} = 35) = (18/37)(1/37)$. Not independent: knowing that red won changes the chance that 14 won.
- Roulette, different spins: the results of separate games are independent.
- The marginal distributions alone do not determine the joint distribution. Two players betting on red and black have the same marginals, but one always loses when the other wins.

Covariance

- Covariance: A measure of how X and Y move together,

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X \mu_Y.$$

- Positive: above-average X tends to come with above-average Y. Negative: they move in opposite directions. Zero: no linear co-movement.
- Roulette: from the joint table,

$$E(y_{Red}\, y_{14}) = (-1)(-1)\tfrac{19}{37} + (1)(-1)\tfrac{17}{37} + (1)(35)\tfrac{1}{37} = \tfrac{37}{37} = 1,$$

$$\mathrm{Cov}(y_{Red}, y_{14}) = 1 - \left(-\tfrac{1}{37}\right)\left(-\tfrac{1}{37}\right) \approx 0.999.$$

- Positive: when 14 wins, red always wins too.

Correlation

- The size of the covariance depends on the units of X and Y. Correlation removes the units:
$$\rho_{XY} = \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}, \qquad -1 \le \rho_{XY} \le 1.$$
- $\rho = 1$ or -1: an exact linear relationship. $\rho = 0$: uncorrelated.
- Roulette: $\sigma_{Red} \approx 1.0$ and $\sigma_{14} \approx 5.84$ (Lecture 5), so $\rho \approx 0.999/(1.0 \times 5.84) \approx 0.17$. A positive but weak relationship.
- If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$.
- The converse fails: if X is symmetric around 0 and $Y = X^2$, then $\mathrm{Cov}(X, Y) = 0$, but Y is completely determined by X. Covariance only measures linear association.

Linear Combinations

- For $W = aX + bY$:

$$E(W) = a\,\mu_X + b\,\mu_Y,$$

$$\mathrm{Var}(W) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\mathrm{Cov}(X, Y).$$

- If X and Y are independent, the covariance term drops out and the variances add.
- If X and Y are (jointly) normal, then W is normal too.
- Roulette: bet \$100 on red and \$10 on 14, so $W = 100\,y_{Red} + 10\,y_{14}$.

$$E(W) = 110 \times (-1/37) \approx -2.97,$$

$$\mathrm{Var}(W) \approx 100^2(1.0) + 10^2(34.08) + 2(100)(10)(0.999) \approx 15{,}399, \qquad \sigma_W \approx 124.$$

Diversification: A Two-Asset Portfolio

Stock: mean 8%, sd 20%. Bond: mean 3%, sd 5%. Correlation 0.2. Portfolio: 60% stock, 40% bond.

- Mean: $0.6(8) + 0.4(3) = 6\%$.
- Covariance: $0.2 \times 20 \times 5 = 20$.
- Variance: $0.6^2(400) + 0.4^2(25) + 2(0.6)(0.4)(20) = 144 + 4 + 9.6 = 157.6$, so sd $\approx 12.6\%$.
- This is less than the weighted average of the sds, $0.6(20) + 0.4(5) = 14\%$.
- Diversification: whenever $\rho < 1$, the portfolio's risk is below the weighted average of the individual risks.
- With n equally weighted assets, the variance falls toward $\rho\sigma^2$, not to zero. Risk shared by all assets (a recession) cannot be diversified away. A sample mean is a portfolio of observations.

Summary

- A continuous random variable has $P(X = x) = 0$. Probabilities are areas under the PDF, $F(x) = P(X \leq x)$, and quantiles invert the CDF.
- Uniform: equally likely values on $[a, b]$. Exponential: waiting times between Poisson events.
- Normal: bell-shaped, $E(X) = \mu$, $\mathrm{Var}(X) = \sigma^2$. Standardize with $Z = (X - \mu)/\sigma$ and use Φ. Work backwards with $x_q = \mu + z_q\sigma$. Remember 1.645 / 1.96 / 2.576.
- The joint distribution gives marginals and conditionals. Independence means the joint PMF is the product of the marginals.
- Covariance and correlation measure linear co-movement.
  $\mathrm{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\mathrm{Cov}(X, Y)$, which explains diversification.

Practice Question 1: Coffee Orders

The time it takes a barista to prepare an order, X (in minutes), is uniformly distributed between 2 and 6 minutes.

(a) Write down the PDF of X.

(b) What is the probability that an order takes more than 5 minutes?

(c) What is $P(3 < X < 4)$? What is $P(X = 4)$?

(d) Find $E(X)$ and the standard deviation of X.

(e) Find the 90th percentile of X.

Practice Question 1: Answers

(a) $f(x) = 1/(6-2) = 1/4$ for $2 \leq x \leq 6$, and 0 otherwise.

(b) $P(X > 5) = (6-5)/4 = 0.25$.

(c) $P(3 < X < 4) = (4-3)/4 = 0.25$. $P(X = 4) = 0$, since X is continuous.

(d) $E(X) = (2+6)/2 = 4$ minutes. $\mathrm{Var}(X) = (6-2)^2/12 = 4/3$, so $\sigma \approx 1.15$ minutes.

(e) $x_{0.9} = 2 + 0.9 \times 4 = 5.6$ minutes.

Practice Question 2: Household Debt

Household debt excluding mortgages, X, is approximately normal with mean $\mu = 41{,}500$ and standard deviation $\sigma = 3{,}900$ (dollars).

(a) What percentage of households have debt above 50,000?

(b) What is the probability that a randomly selected household has debt between 35,000 and 45,000?

(c) What level of debt separates the top 20% of households from the rest?

(d) A household is at the 65th percentile of debt. How much debt does it have?

Practice Question 2: Answers

(a) $z = (50000 - 41500)/3900 \approx 2.18$, so
$P(X > 50000) = 1 - \Phi(2.18) \approx 1 - 0.9854 = 0.0146$. About 1.5%.

(b) $z = (35000 - 41500)/3900 \approx -1.67$ and $z = (45000 - 41500)/3900 \approx 0.90$, so

$$P(35000 < X < 45000) = \Phi(0.90) - \Phi(-1.67) \approx 0.8159 - 0.0475 \approx 0.77.$$

(c) The 80th percentile: $z_{0.80} \approx 0.84$, so $x_{0.80} = 41500 + 0.84 \times 3900 \approx 44{,}780$.

(d) $z_{0.65} \approx 0.385$, so $x_{0.65} = 41500 + 0.385 \times 3900 \approx 43{,}000$.

Practice Question 3 Two Projects

A firm can invest in two projects. Project A has an expected return of 10% with standard deviation 4%. Project B has an expected return of 14% with standard deviation 8%. The correlation between the returns is $\rho = -0.3$.

(a) Find the covariance of the two returns.

(b) The firm splits its money equally, so its return is $W = 0.5A + 0.5B$. Find $E(W)$ and the standard deviation of W.

(c) Compare the portfolio with investing only in A. Which has the higher mean? Which has the lower risk?

(d) Assume the returns are jointly normal. What is the probability that the portfolio return is negative?

Practice Question 3: Answers

(a) $\mathrm{Cov}(A, B) = \rho\,\sigma_A\sigma_B = -0.3 \times 4 \times 8 = -9.6$.

(b) $E(W) = 0.5(10) + 0.5(14) = 12\%$.

$$\mathrm{Var}(W) = 0.25(16)+0.25(64)+2(0.5)(0.5)(-9.6) = 4+16-4.8 = 15.2, \qquad \sigma_W \approx 3.9\%.$$

(c) The portfolio has a higher mean (12% vs. 10%) and slightly lower risk (3.9% vs. 4%) than A alone. The negative correlation makes diversification especially effective.

(d) $W \sim N(12, 3.9^2)$, so $P(W < 0) = \Phi\left(\frac{0-12}{3.9}\right) = \Phi(-3.08) \approx 0.001$.